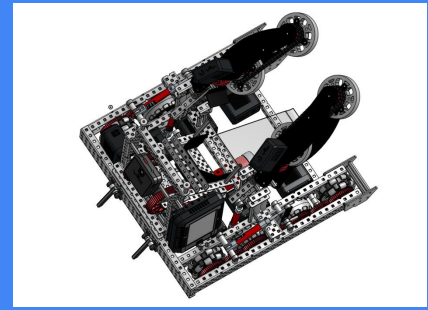


CNN and Computer Vision Techniques for Game Element and Robot Detection

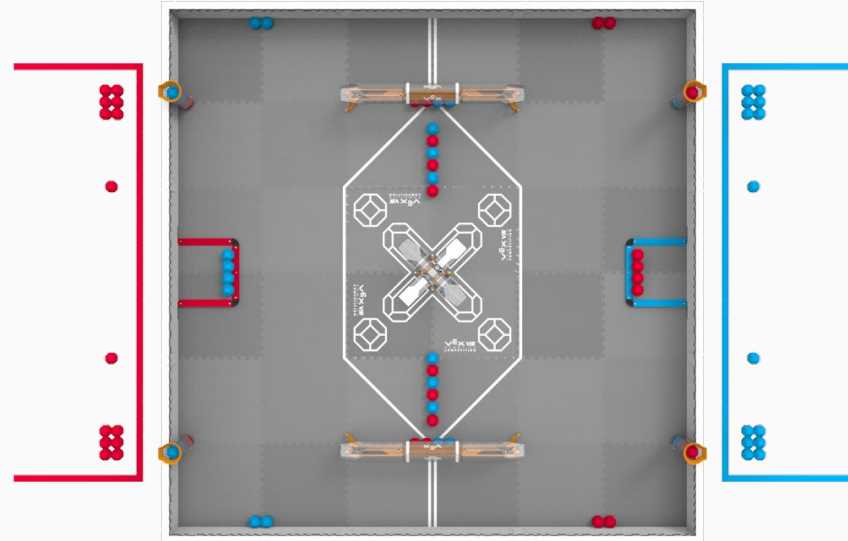
Jason Asariah, Titus Rehard, Hannah Gray



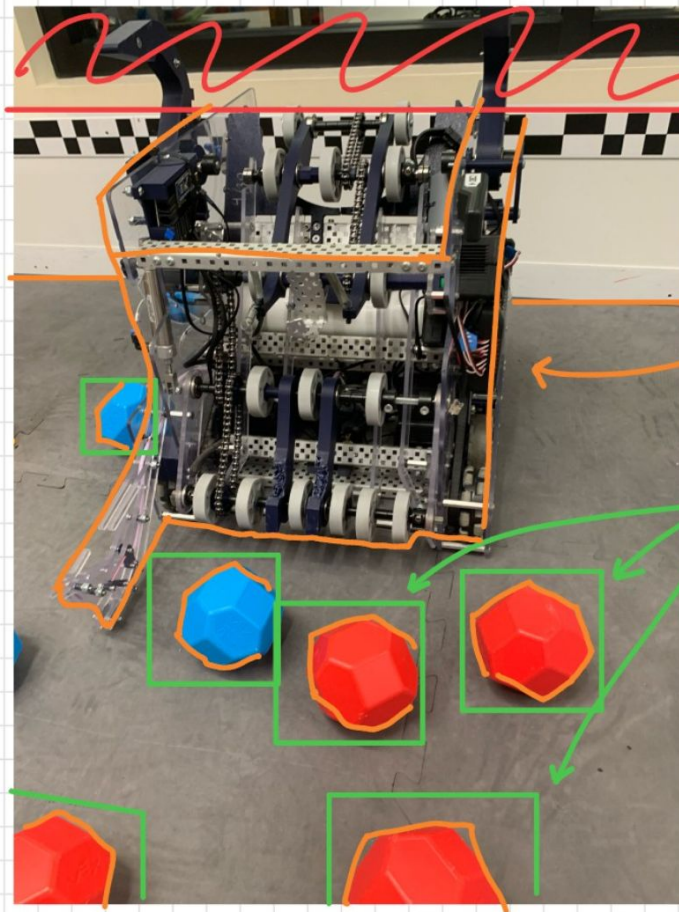
Background



- During the VEXU autonomous period, a computer vision system should be able to locate game elements as well as the opponents' robots
- Identifying these robots will allow us to:
 - 1) Avoid hitting any other robots
 - 2) Analyze and adapt to their strategy
- **Robots can look very different**



Strategy



remove

edge detection

ID all
game
elements

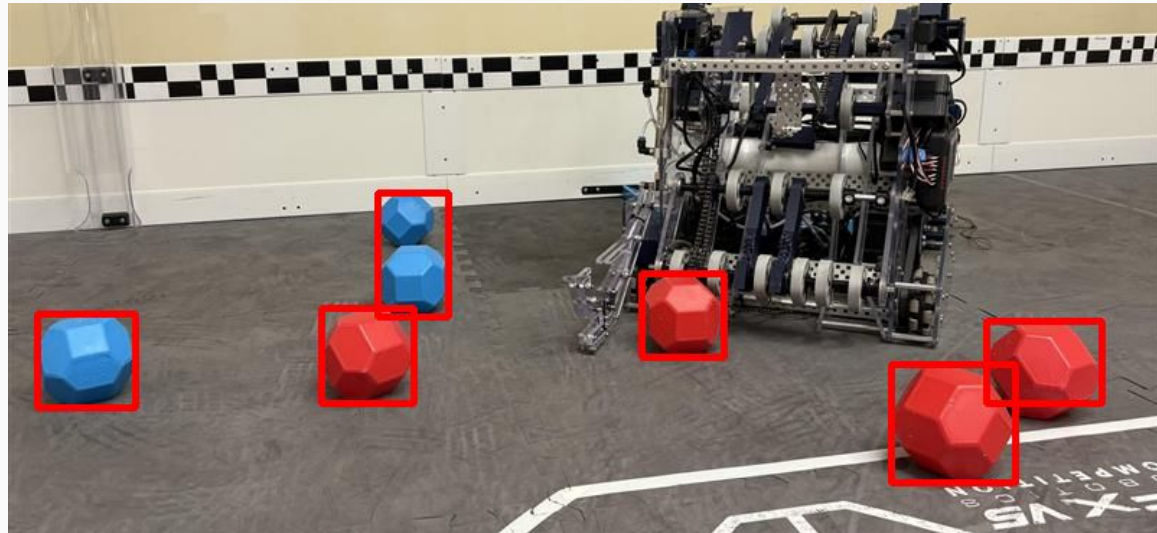
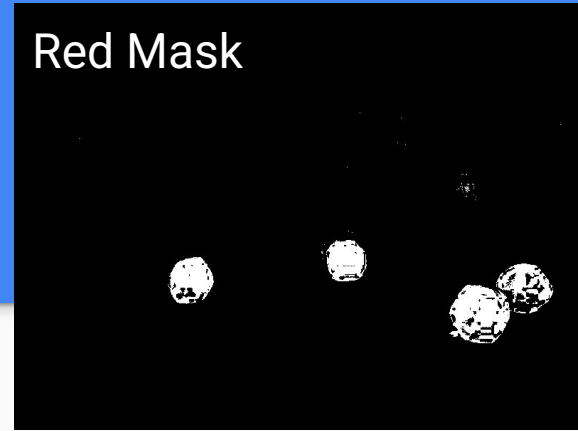
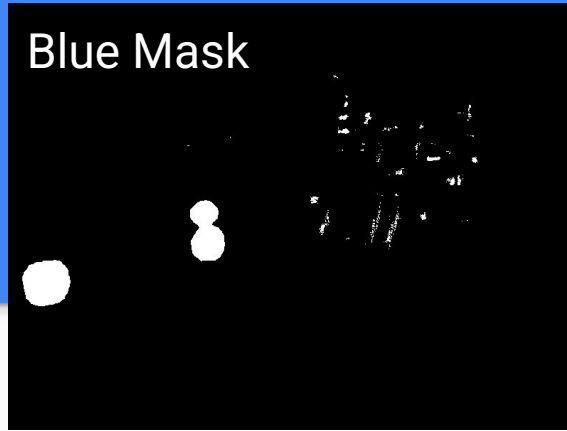
Color Detection

Steps:

- Blue / Red Mask
- Detect Large Contours
- Box

Weaknesses:

- Light Differences
- Contours Connected



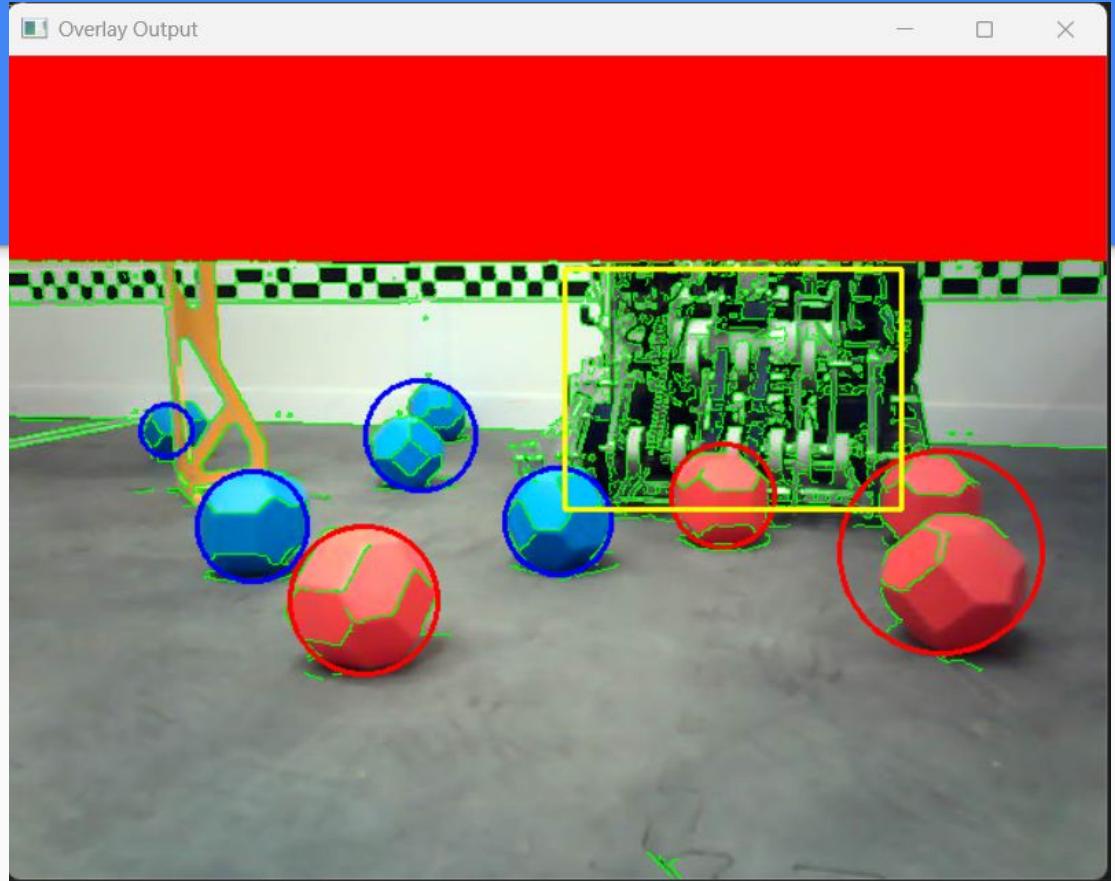
Edge Detection

Blocks:

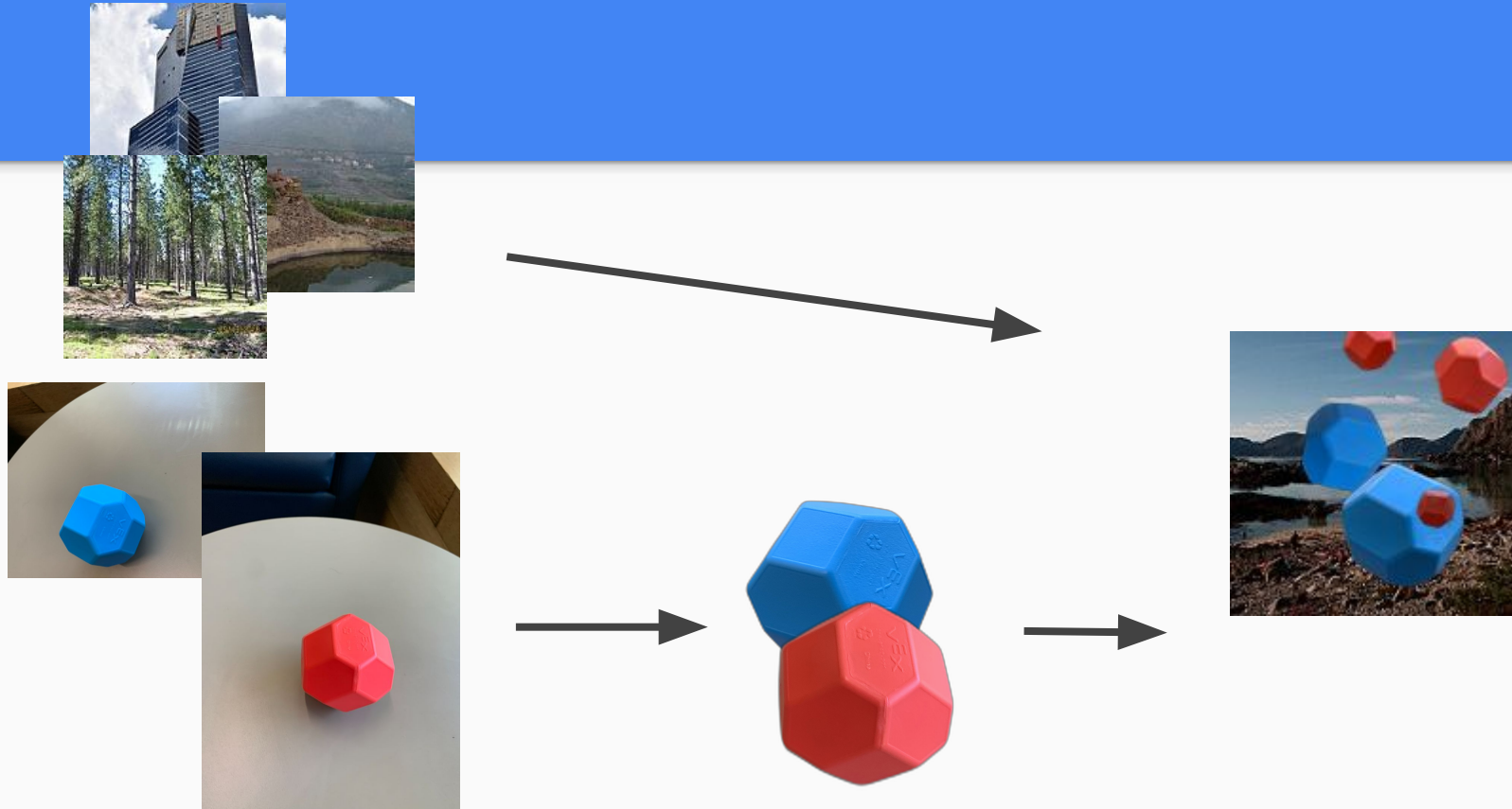
- Color Filtering
- Edge Detection
- Approximated Circle Overlay

Robot (Interesting Entity):

- Edge Detection
- Edge Cluster Boxing

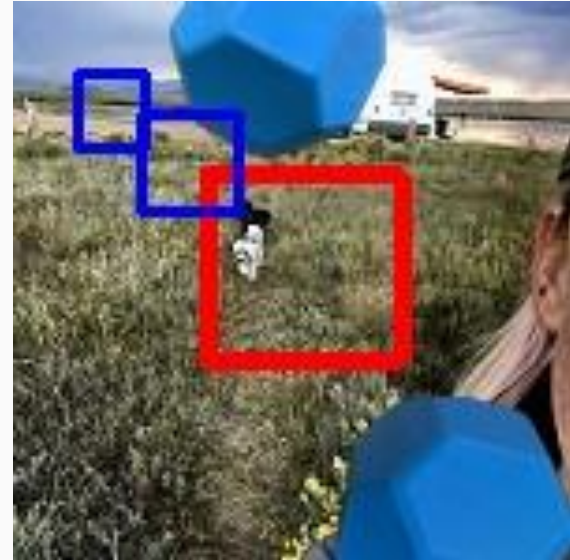
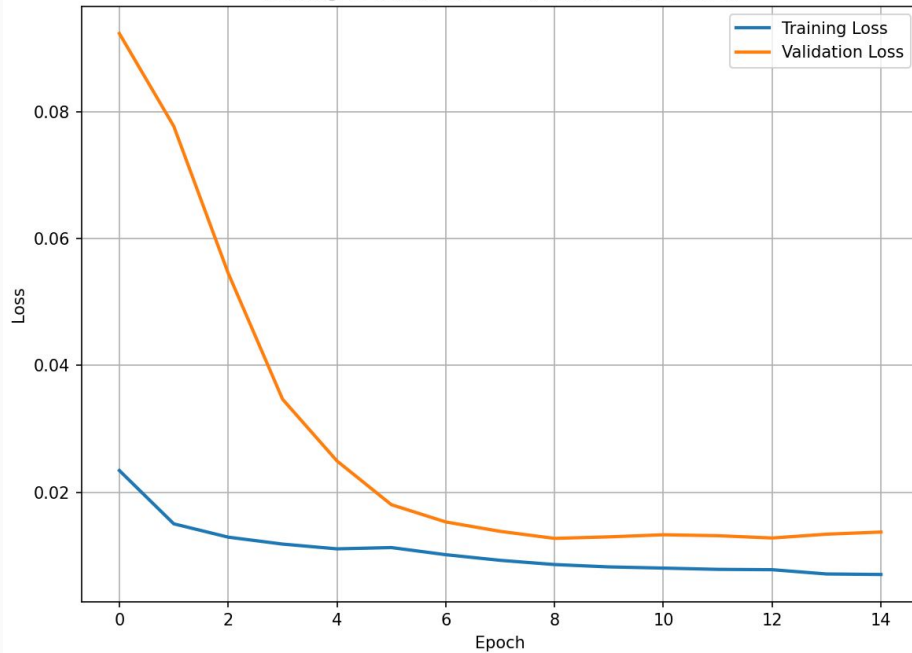


CNN - Synthetic Data Generation

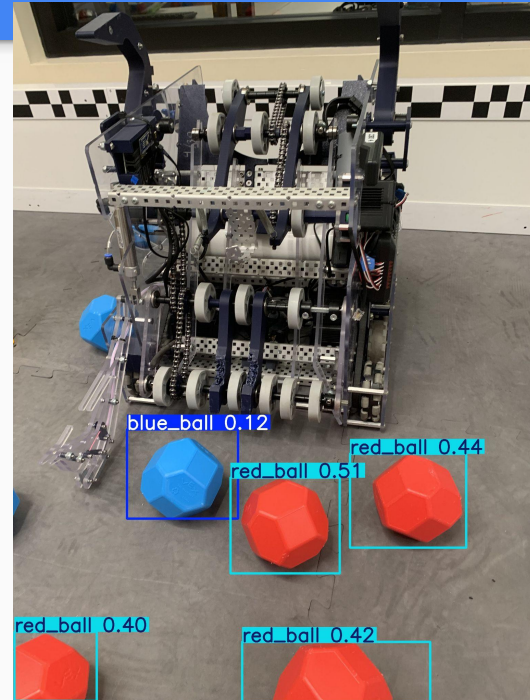


CNN - Model Training/Architecture

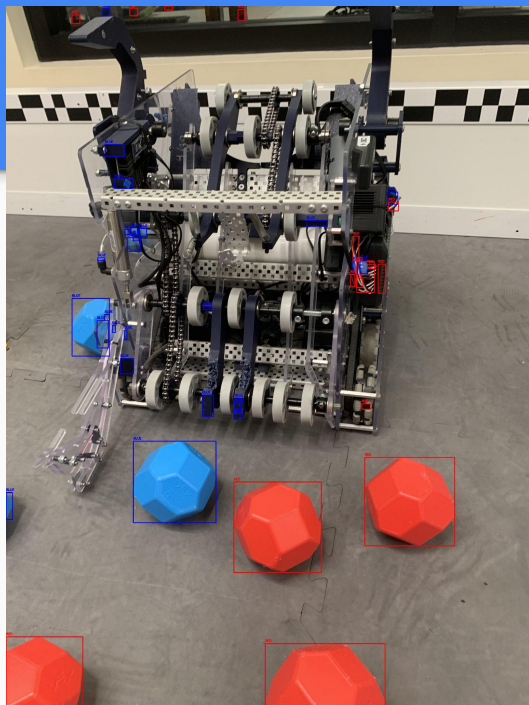
Training vs Validation Loss (Huber, MobileNetV2)



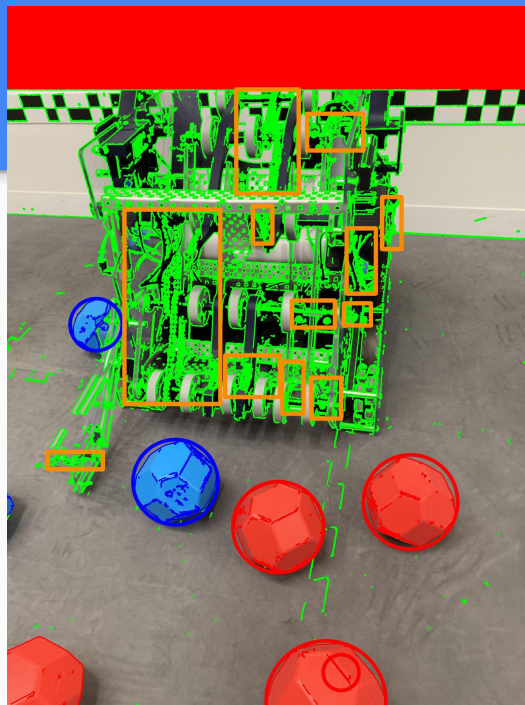
CNN - Model Training/Architecture



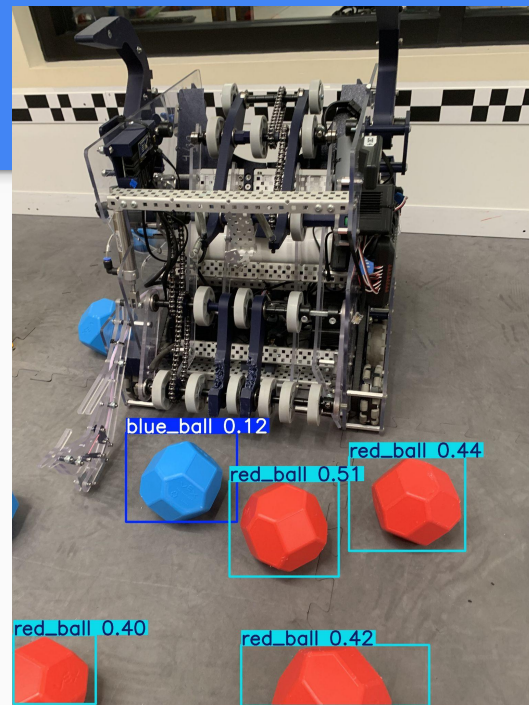
Integration



Color Detection
Finds game elements



Edge Detection
Finds robot and other
element edges



CNN
Finds game elements

Conclusion

- Able to detect red and blue game elements using each method
- Edge Detection method works decently to detect robot (the other two don't attempt to detect the robot due to not knowing how it looks)
- Goals and barriers can conflict with color detection as well as edge cluster detection